Miguel E. Rodriguez Martinez

Senior Full-Stack Engineer | Real-Time 3D & AI Data Platforms

miguelxlerion@gmail.com | +57 313 226 3265 | Bogotá, Colombia
linkedin.com/in/miguelrodriguez-dataviz | mike-portfolio.xlerionultimate.workers.dev
github.com/miguelxlerion

Professional summary

Senior Full-Stack Engineer with 10+ years building real-time data platforms, digital twins and interactive 3D visualization. Scalable full-stack architecture (React, Node.js, Python/Django), vector databases (pgvector) and IoT pipelines. Proven record: 60–80% faster analysis times, 99% uptime on mission-critical systems and code tested with CI/CD.

Professional Experience

Senior Full-Stack Engineer (Sole Technical Owner)

Caucasia Mayor's Office — FOVIS | Apr. 2024 – Oct. 2025

- Architected and built solo a scalable platform for the government property management system (12+ municipalities, 1,200+ housing units) with React 18, Tailwind and Node.js/Express.
- Designed and implemented real-time KPI dashboards (WebSockets) for complex data visualization, cutting administrative decision-making time by 60%.
- Built a React component library with testing (Jest + React Testing Library) and Storybook, ensuring UX consistency and responsive design.
- Set up CI/CD pipelines (GitHub Actions) with linting, testing and automated deployment to Docker/Linux, achieving 99% uptime in production through fault-tolerant design and local buffering.

Senior Full-Stack Engineer & 3D Interactive Lead

Xlerion Studio | Feb. 2023 – Present

- Led the integration of complex frontend architectures connecting real-time data streams (WebSockets/MQTT) with digital twins and interactive 3D scenes (Three.js, React Three Fiber).
- Optimized visual performance, achieving stable FPS through profiling, lazy loading and level-of-detail (LOD) techniques.
- Built a MaxScript/Python automation pipeline that reduced 3D asset export time by 80% and eliminated manual errors.
- Implemented E2E (Cypress) and unit (Jest/PyTest) testing in GitHub Actions/GitLab CI pipelines, with 85%+ coverage.

Front-End Engineer & Data Visualization Lead

AONC | Mar. 2018 – Dec. 2019

- Led the development of gamified interfaces and 3D visualizations (Three.js/WebGL) that turned massive datasets into navigable environments, reducing analysis time by 40%.
- Established code standards, code reviews and automated testing (Jest, PyTest), improving codebase maintainability.

Front-End Engineer & UI Designer

USAID — ACDI/VOCA | Jan. 2015 – Dec. 2017

- Architected the ICOE software, translating complex national databases into interactive dashboards for government decision-making.
- Ensured accessibility (WCAG AA) and responsive design for senior stakeholders, and migrated the legacy system to a component-based stack.

Projects

Enterprise RAG Assistant

Modular backend (Python/Django/DRF) connecting corporate knowledge bases to LLMs through RAG: asynchronous PDF processing, semantic chunking, hybrid search (vector + lexical) and re-ranking to prevent hallucinations. Stack: LangChain, PostgreSQL + pgvector, Celery/Redis, Docker; tested with PyTest and CI/CD on GitHub Actions.

IoT Resilient Telemetry Backend

Real-time ingestion and processing platform for high-volume IoT data streams (Python asyncio / Node.js, WebSockets and MQTT). Achieved 99% uptime through fault-tolerant design, local buffering and load testing with Locust. Stack: PostgreSQL/MySQL, Docker, Linux and Bash.

Interactive 3D Analytics Dashboard

Full-stack application (React, Three.js, Django/Node.js) that turns massive tabular data into real-time 3D dashboards and gamified environments, with stable FPS through LOD and lazy loading. Tested with Jest/Cypress and automated CI/CD.

Education

AI Bootcamp — Explorer Level (159 h)

MinTIC & Universidad Sergio Arboleda | 2024

Diploma in Introduction to Local Public Management (80 h)

Fundación de Educación Superior San José & Gobernación de Cundinamarca | 2025

Skills

Frontend: React 18, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, React Three Fiber, Three.js, WebGL

Backend: Node.js/Express, Python (Django/DRF, FastAPI), PostgreSQL, MongoDB, Redis, WebSockets, MQTT

DevOps & Quality: Docker, Docker Compose, GitHub Actions, GitLab CI, Linux (Ubuntu Server), Nginx, Jest, React Testing Library, Cypress, PyTest

3D / AI / Data: Three.js, WebGL, Digital Twins, Blender, MaxScript (automation), RAG/LangChain, pgvector (Vector Databases), semantic chunking, hybrid search

Certifications

Master Power BI (4 h) — Intelligent Training · MinTIC | 2024

Introduction to Unreal Engine — SAE Institute México / Coursera | 2024

Software Project Estimation (5 h) — Udemy | 2020

Strategic Excellence and Organizational Adaptation (20 h) — SENA & Cámara Oficial de Comercio de España | 2026

ChatGPT Fundamentals and Integration (2 h) — HPE Education Services | 2026

Third-Person Video Game Design with Unity (40 h) — SENA | 2015

Languages

Spanish — Native · English — B2